Finding the Right Law, in Khmer

Comparing Four Deep-Learning Retrievers on Cambodian Statutory Law & Citation-Grounded Assistant

Deep Learning Final Project (Individual Assessment — 50%) · Mengchheang Long
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10-Minute Presentation + 5-Minute Q&A (Course Syllabus §6.2 & §6.3)

1,976	200	4 Architectures	+44.1%
Khmer statutory articles (100% consecutive)	Real human legal test questions	PyTorch DL models (A1–A4)	MRR@10 gain vs. keyword search (p < 0.001)

PRESENTATION

⏱ 0:20 – 1:10 (50s)

1. Problem Definition & Practical Motivation

A citizen asks a question in plain Khmer — can deep learning retrieve the exact statutory article out of 1,976?

Task Definition (§5.A)

Input: Unstructured legal question in natural Khmer (q).

Output: Top-k ranked statutory articles $\{d_1, \dots, d_k\}$ from the full 1,976-article corpus without statute/chapter filter hints.

«តើកិច្ចសន្យាបង្កើតឡើងដោយរបៀបណា?»

("How is a contract formed under the Civil Code?")

→ Ground Truth: **Civil Code Art. 336** (ការបង្កើតកិច្ចសន្យា...) & **Art. 311** (និយមន័យនៃកិច្ចសន្យា)

Why Lexical Keyword Search (BM25) Fails

- **Severe Vocabulary Mismatch (57.5% of BM25 misses):** Citizens ask in conversational phrasing; statutes use formal civil law terminology.
- **Misleading Term Distraction:** BM25 matches common query words like "របៀប" (manner) and retrieves irrelevant procedural articles (e.g., Art. 623 "Manner of dividing proceeds") while completely missing Art. 336.
- **Dense Semantic Representation:** Neural bi-encoders map queries and statutes into a shared latent geometry, bridging vocabulary discrepancies via contextual meaning.

PRESENTATION

⏱ 1:10 – 2:10 (60s)

2. Dataset & Preprocessing: The Limon Font Challenge

Official government PDFs were typed in legacy 1990s ASCII fonts — extraction had to be engineered from scratch

The Legacy Limon Problem

- **Corpus Source:** Official PDFs from Open Development Cambodia (Civil Code 2007: 1,304 articles, Criminal Code 2009: 672 articles).
- **Obsolete ASCII Encoding:** Text layer contains raw Latin keystrokes (e.g. `maRta 336>-karbegáItkic©snñ...`). Standard PDF tools produce unreadable gibberish.
- **Visual Grapheme Inversion:** Khmer pre-base vowels (ើ ៀ ែ) and subscript consonants are typed *before* the base consonant visually, violating Unicode logical character ordering.

Engineered Solution & Validation

- **Custom Pipeline:** Engineered `LimonPdfExtractor` with byte-level font mapping and a finite-state machine for syllable reordering into Unicode.
- **100% Consecutive Extraction:** Cleanly extracted all 1,976 articles (Civil 1–1304, Criminal 1–672) with zero missing provisions, audited against physical gazettes.
- **Structural Metadata:** Preserved full hierarchical breadcrumbs (Book → Title → Chapter → Section → Article Number).

PRESENTATION

⏱ 2:10 – 3:00 (50s)

3. Training, Validation & Test Splits (Zero Leakage)

Rigorous split quarantine and data sanitization protocol enforced across all models (Seed 42)

Weak Supervision Data Generation

- **Title-to-Body Pairs:** Because labeled Khmer legal QA datasets do not exist, 1,472 unique article titles served as natural query-body training pairs.
- **Ambiguous Titles Excluded:** 270 articles sharing identical titles (e.g., "Definitions", "Competence") were quarantined to prevent ambiguous label noise.
- **Passage Title Stripping:** Titles were strictly removed from passages during training to prevent trivial lexical copy-matching.

Benchmark Splits & Leakage Guard (§5.A)

- **Fixed Splits (Seed 42):** 1,176 Train (80%) · 148 Validation (10%) · 148 Held-Out Test Titles (T-T Secondary Benchmark).
- **Primary T-Q Benchmark:** 200 authentic human legal questions (100 Civil + 100 Criminal).
- **Zero Test Leakage:** All 200 questions and their 261 relevant statutory articles were strictly quarantined from training. Test sets evaluated strictly once.

PRESENTATION

⏱ 3:00 – 4:15 (75s)

4. Model Architectures & Strategies

Comparing 4 distinct PyTorch deep learning approaches across model families and training strategies (§4)

Approach	Encoder Architecture (Dim. A)	Training Strategy (Dim. B)	Trainable Params	Tokenization
BM25	Sparse lexical inverted index (Baseline)	Frequency heuristics ($k_1=1.5, b=0.75$)	0	khmer-nltk CRF
E5 Zero-Shot	Pretrained multilingual transformer	Zero-shot inference (Frozen)	0 (278M)	SentencePiece
A1 BiLSTM	Bidirectional LSTM (2 layers, hidden 256)	Trained from scratch (Random init)	1.97 M	khmer-nltk CRF
A2 Linear Probe	multilingual-e5-base (XLM-RoBERTa)	Frozen backbone + trained linear head	0.59 M	SentencePiece
A3 Full Fine-Tune ★	**multilingual-e5-base (XLM-RoBERTa)**	**Full fine-tuning (All 12 layers adapted)**	**278.04 M**	**SentencePiece**
A4 PrahokBART	prahokbart-base (Native Khmer seq2seq)	Full fine-tuning with mean-pooling	35.83 M	SentencePiece (32k)

Course Rubric Compliance (§4): Approaches differ in **architecture** (BiLSTM vs Transformer vs BART) AND **training strategy** (scratch vs linear probe vs fine-tuning), exceeding the 3-approach requirement.

PRESENTATION

⏱ 4:15 – 5:15 (60s)

5. Training Process & InfoNCE Contrastive Loss

Shared contrastive learning formulation with in-batch negatives and symmetric cross-entropy

Symmetric InfoNCE Objective

Both query-to-document and document-to-query contrastive losses are averaged across batch size B:

```
L = -1/(2B) * Σ [ log( e^(sim(q_i,d_i)/τ) / Σ_j e^(sim(q_i,d_j)/τ) ) + log( e^(sim(d_i,q_i)/τ) / Σ_j e^(sim(d_j,q_i)/τ) ) ]
```

- **Temperature $\tau = 0.05$:** Sharply penalizes hard negative false positives.
- **In-Batch Negatives:** $B-1$ negative articles per sample without mining overhead.

Optimization & Engineering Controls

- **Optimizer:** AdamW with linear warmup (10% of total steps) and cosine learning rate decay.
- **Regularization:** Weight decay (0.01) + embedding dropout + contrastive repulsion.
- **Checkpointing (§5.B):** `torch.save` saves model weights, optimizer, scheduler, and RNG state each epoch (`last.pt`, `best.pt`) with `--resume` flag support.

PRESENTATION

⏱ 5:15 – 6:10 (55s)

6. Experimental Setup & Hyperparameter Tuning

Systematic hyperparameter search recorded across learning rates and weight decay on Tesla T4 GPU (§5.C)

Approach A3: 6-Run Grid Search (Tesla T4)

Run Name	LR	WD	Best Epoch	Val MRR@10
a3_lr2e-5_wd0.01 ★	**2e-5**	**0.01**	**4**	**0.9095**
a3_lr2e-5_wd0.0	2e-5	0.00	4	0.9095
a3_lr3e-5_wd0.0	3e-5	0.00	4	0.9082
a3_lr3e-5_wd0.01	3e-5	0.01	4	0.9081
a3_lr1e-5_wd0.01	1e-5	0.01	3	0.9023
a3_lr1e-5_wd0.0	1e-5	0.00	3	0.9023

Controlled Environments & Other Models

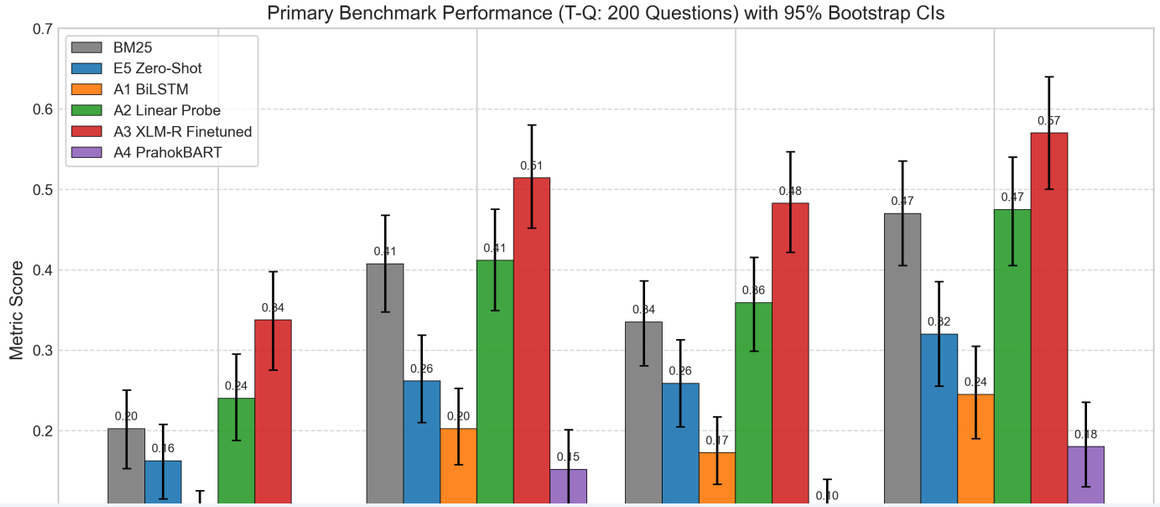
- **A1 BiLSTM Tuning:** LR {1e-3, 3e-3} × Dropout {0.1, 0.3} → Best: LR 3e-3, Drop 0.3 (Val MRR: 0.7679).
- **A2 Linear Probe Tuning:** LR {1e-3, 1e-2} × WD {0, 0.01} → Best: LR 1e-3, WD 0.01 (Val MRR: 0.7894).
- **A4 PrahokBART Tuning:** LR {5e-5, 1e-4} × WD {0.01} → Best: LR 1e-4 (Val MRR: 0.6116).
- **Reproducibility:** Global seed 42 fixed across Python, NumPy, PyTorch, cuDNN.

PRESENTATION

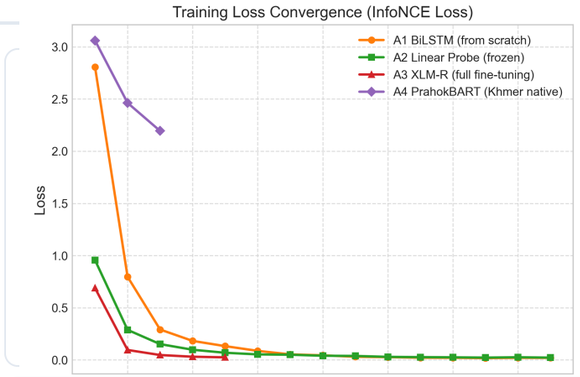
⏱ 6:10 – 7:25 (75s)

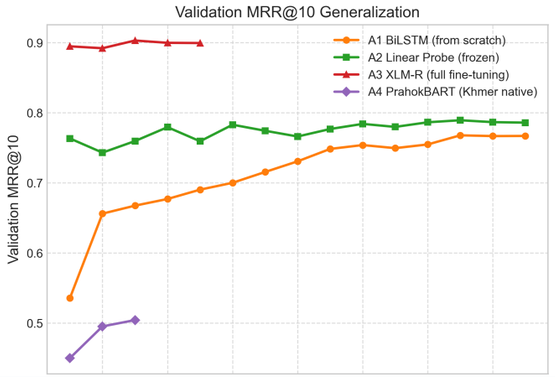
7

E …fer (§5.C)



Model	Recall@1	Recall@5	MRR@10	Hit@5
A3 XLM-R Fine-Tuned ★	**0.3375**	**0.5142**	**0.4827**	**57.0%**
A2 XLM-R Linear Probe	0.2400	0.4117	0.3591	47.5%
BM25 Baseline	0.2025	0.4075	0.3350	47.0%
E5 Zero-Shot	0.1625	0.2617	0.2588	32.0%
A1 BiLSTM	0.0875	0.2025	0.1724	24.5%
A4 PrahokBART	0.0525	0.1517	0.1035	18.0%





A1 (scratch): Overfits severely (train loss → 0.02, val stalls). With only ~1.1k pairs, random embeddings fail to capture Khmer legal morphology.

A2 (frozen): Cannot overfit, but linear projection head hits an expressive capacity ceiling (ties BM25 at p = 0.9031).

A3 (fine-tuned): Adapts all 12 transformer layers, achieving statistically validated superiority (+44.1% MRR@10, p < 0.001).

PRESENTATION

⏱ 7:25 – 8:20 (55s)

8. Qualitative Error Analysis & Study Limitations

Categorizing failure modes on the 200 benchmark questions and measuring distribution shift (§5.C)

Error Taxonomy at k=5 (T-Q: 200 Questions)

Model	Misses	Hit@5	Vocab Mismatch	Multi-Article	Near-Miss
A3 XLM-R	**86**	**57.0%**	**52 (60.5%)**	**15 (17.4%)**	**17 (19.8%)**
A2 Linear Probe	105	47.5%	66 (62.9%)	27 (25.7%)	7 (6.7%)
BM25 Baseline	106	47.0%	61 (57.5%)	26 (24.5%)	15 (14.2%)
A1 BiLSTM	151	24.5%	79 (52.3%)	37 (24.5%)	19 (12.6%)
A4 PrahokBART	164	18.0%	96 (58.5%)	44 (26.8%)	8 (4.9%)

Near-misses represent relevant articles correctly placed at rank 6–10.

Distribution Shift & Limitations (§5.D)

- **In-Distribution (T-T: 97–99% Recall@5):** On held-out title-to-body retrieval, performance is near ceiling because training pairs shared structural alignment.
- **Out-of-Distribution (T-Q: 15–51% Recall@5):** Human questions cause severe distribution shift due to colloquial phrasing, scenario narratives, and synonym variations.
- **Study Limitations:** Evaluated on Civil and Criminal Codes (1,976 articles); scanned labor law PDFs requiring OCR left for future work.

PROTOTYPE & LIVE DEMO

⏱ 8:20 – 9:20 (60s Live Demo)

9. Citation-Grounded Prototype Demonstration

End-to-end RAG assistant running on localhost:8501 (PyTorch A3 Retriever + FastAPI + Streamlit + DeepSeek Flash)

1. CITIZEN QUERY
តើកិច្ចសន្យាបង្កើតឡើង...
→
2. A3 DENSE ENCODER
768-d Vector (<100ms)
→
3. 1,976 ARTICLES
Cosine Similarity Ranking
→
4. CITATION GUARD
Verbatim Statute Check
→
5. DEEPSEEK FLASH
Khmer Legal Synthesis

Resolving The Semantic Mismatch

"តើកិច្ចសន្យាបង្កើតឡើងដោយរបៀបណា យោងតាមក្រមរដ្ឋប្បវេណី?"

BM25 matches "របៀប" → Art. 623 (Miss). A3 Neural maps semantic intent → **Art. 311** (Rank 1, sim 0.61) & **Art. 336** (Rank 2, sim 0.53)!

Anti-Hallucination Citation Guard

Generates grounded Khmer analysis with automated statutory verification tags matched against retrieved legal chunks:

✓ Civil Code 2007 — Article 336 [Verified Citation]

Live Demonstration at localhost:8501

Streamlit UI running full PyTorch A3 encoder and automated benchmark comparison dashboard

http://localhost:8501

CONCLUSION

⏱ 9:20 – 10:00 (40s)

10. Conclusions & Defense Summary

Three key takeaways, engineering rigor, and transition to Q&A (§6.2)

1. **Pre-training is Indispensable:** Models trained from scratch (BiLSTM) fail outright on small legal data (0.1724 MRR@10) due to vocabulary sparsity; multilingual pretraining provides necessary structural priors.

2. **Full Fine-Tuning is Decisive:** Adapting all transformer layers (A3) is strictly required to surpass keyword search on conversational legal queries (+44.1% relative MRR@10 over BM25, $p < 0.001$).

3. **Reproducible Engineering:** 100% consecutive Limon extraction (1,976 articles), zero test leakage, 136 passing automated unit tests, and GPL-3.0 open-source code.

Thank you for your attention. I welcome your questions.

Appendix Slides A1–A4 Ready for 5-Min Q&A

A1. Paired Bootstrap Hypothesis Testing

10,000 bootstrap resamples on paired per-query score differences (Primary Benchmark T-Q: 200 Questions)

Comparison	Metric	Mean A	Mean B	Difference (Δ) [95% CI]	p-value	Wins / Losses / Ties
A3 vs BM25 Baseline	Recall@5	0.5142	0.4075	+0.1067 [+0.0500, +0.1633]	**0.0002**	41W / 13L / 146T
A3 vs BM25 Baseline	MRR@10	0.4827	0.3350	+0.1477 [+0.0962, +0.1993]	**0.0001**	72W / 25L / 103T
A3 vs E5 Zero-Shot	MRR@10	0.4827	0.2588	+0.2239 [+0.1685, +0.2787]	0.0001	80W / 15L / 105T
A3 vs A1 BiLSTM	MRR@10	0.4827	0.1724	+0.3103 [+0.2483, +0.3705]	0.0001	104W / 13L / 83T
A3 vs A2 Linear Probe	MRR@10	0.4827	0.3591	+0.1236 [+0.0796, +0.1697]	0.0001	65W / 21L / 114T
A2 vs BM25 Baseline	Recall@5	0.4117	0.4075	+0.0042 [-0.0567, +0.0625]	**0.9031 (n.s.)**	26W / 25L / 149T
A3 vs A4 PrahokBART	MRR@10	0.4827	0.1035	+0.3792 [+0.3195, +0.4396]	0.0001	116W / 6L / 78T
A4 vs BM25 Baseline	MRR@10	0.1035	0.3350	-0.2315 [-0.2869, -0.1775]	0.0001	12W / 90L / 98T

Interpretation: A3's gain over BM25 is statistically significant ($p < 0.001$). A2 vs BM25 ($p = 0.9031$) proves statistical parity: frozen representations match keyword search but require fine-tuning to surpass it.

A2. Complete Empirical Benchmark Leaderboard

Primary (T-Q: 200 Questions) & Secondary (T-T: 148 Titles) with 95% Bootstrap Confidence Intervals

Primary Benchmark (T-Q: 200 Human Legal Questions)

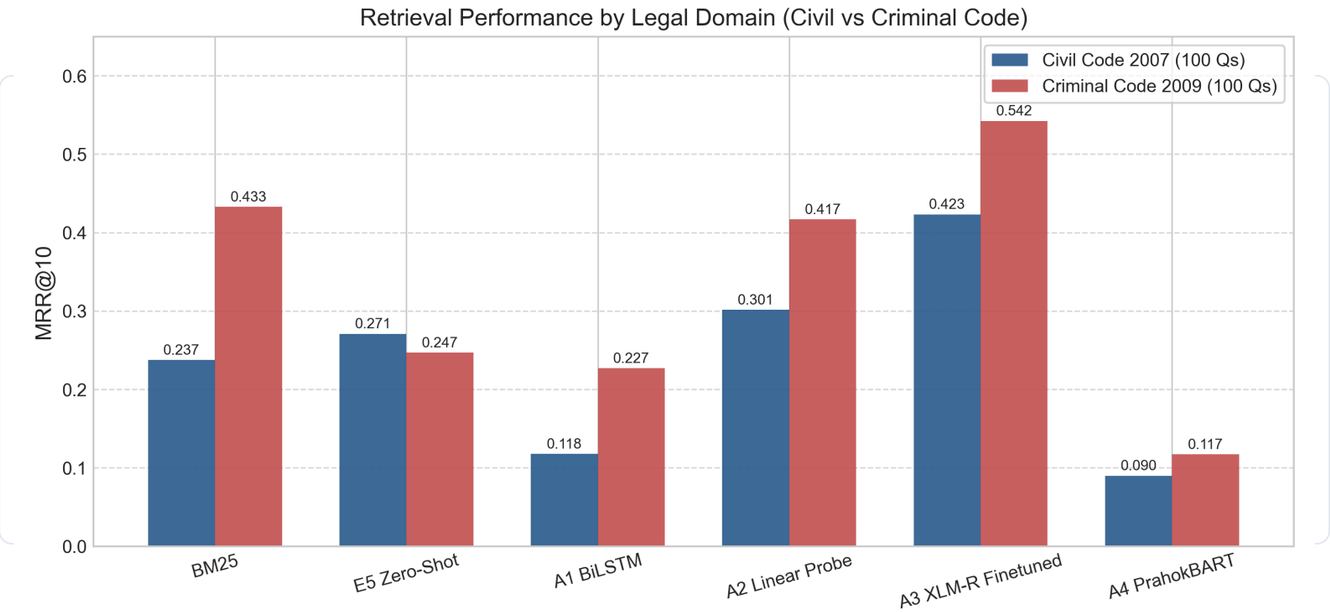
Model	Recall@1 [95% CI]	Recall@5 [95% CI]	MRR@10 [95% CI]	Hit@5 [95% CI]	Params	Hardware
A3 XLM-R Full FT ★	**0.3375 [0.28, 0.40]**	**0.5142 [0.45, 0.58]**	**0.4827 [0.42, 0.55]**	**57.0% [50, 64]**	**278.04M**	**GPU (T4, fp32)**
A2 XLM-R Linear Probe	0.2400 [0.19, 0.30]	0.4117 [0.35, 0.48]	0.3591 [0.30, 0.42]	47.5% [41, 54]	0.59M	CPU (5.9s)
BM25 Baseline	0.2025 [0.15, 0.25]	0.4075 [0.35, 0.47]	0.3350 [0.28, 0.39]	47.0% [41, 54]	0	CPU (4ms)
E5 Zero-Shot	0.1625 [0.12, 0.21]	0.2617 [0.21, 0.32]	0.2588 [0.20, 0.31]	32.0% [26, 39]	0	CPU
A1 BiLSTM (scratch)	0.0875 [0.05, 0.13]	0.2025 [0.16, 0.25]	0.1724 [0.13, 0.22]	24.5% [19, 31]	1.97M	CPU (2,035s)
A4 PrahokBART	0.0525 [0.03, 0.09]	0.1517 [0.11, 0.20]	0.1035 [0.07, 0.14]	18.0% [13, 24]	35.83M	CPU (1,602s)

Secondary Benchmark (T-T: 148 Held-Out Article Titles)

Model	Recall@1 [95% CI]	Recall@5 [95% CI]	MRR@10 [95% CI]	Hit@5 [95% CI]
A3 XLM-R Full FT ★	**0.9054 [0.85, 0.95]**	**0.9932 [0.97, 1.00]**	**0.9456 [0.92, 0.97]**	**99.3% [97, 100]**
BM25 Baseline	0.7973 [0.74, 0.86]	0.9730 [0.95, 0.99]	0.8677 [0.82, 0.91]	97.3% [95, 99]
A2 XLM-R Linear Probe	0.7905 [0.72, 0.85]	0.9392 [0.90, 0.97]	0.8563 [0.81, 0.90]	93.9% [90, 97]
A4 PrahokBART	0.5946 [0.51, 0.68]	0.8041 [0.74, 0.87]	0.6733 [0.60, 0.74]	80.4% [74, 87]
A1 BiLSTM (scratch)	0.5135 [0.43, 0.60]	0.7568 [0.70, 0.82]	0.6242 [0.56, 0.69]	75.7% [70, 82]
E5 Zero-Shot	0.5405 [0.46, 0.61]	0.7230 [0.65, 0.78]	0.6235 [0.55, 0.69]	72.3% [65, 78]

A3. Per-Code Breakdown (Civil Code vs Criminal Code)

Evaluating performance separately on Civil Code (100 questions) and Criminal Code (100 questions)



Model	Civil R@5	Civil MRR	Criminal R@5	Criminal MRR
A3 XLM-R	**45.8%**	**0.4232**	**57.0%**	**0.5422**
BM25 Baseline	30.5%	0.2374	51.0%	0.4327
A2 Linear Probe	37.0%	0.3015	45.3%	0.4167
A1 BiLSTM	13.0%	0.1177	27.5%	0.2270
A4 PrahokBART	13.3%	0.0898	17.0%	0.1173

Key Insight: Criminal Code performance is systematically higher across models (+10% to +15% MRR) because criminal offenses have sharper conceptual boundaries (theft, homicide, forgery) compared to abstract civil contracts.

A4. Concrete Qualitative Failure Cases (from results/error_analysis/)

Four verified ground-truth query walkthroughs illustrating specific linguistic failure modes

1. Sub-Clause Specificity (A3 Rank 6)

«តើអ្នកលក់ត្រូវទទួលខុសត្រូវលើវិការៈកំបាំងនៃវត្ថុទិញដូរយ៉ាងដូចម្តេច?»

Latent defect in sales contract. Expected: Civil Code Art. 544.

A3 retrieves Art. 547 (Remedies for defect) at Rank 1, and Art. 544 at Rank 6 (near-miss).

2. Multi-Article Query Coverage

«តើប្រភេទនៃសិទ្ធិប្រត្យក្សមានអ្វីខ្លះនៅក្នុងក្រមរដ្ឋប្បវេណី?»

Classes of real rights. Expected: Civil Code Art. 130 & 131.

BM25 catches only 1 in top-5; A3 catches **both** at ranks 2 and 3.

3. Code Confusion (A1 Failure)

«តើច្បាប់ហាមឃាត់ការប្រើប្រាស់សិទ្ធិដោយបំពានយ៉ាងដូចម្តេច?»

Abuse of rights. Expected: Civil Code Art. 4.

A1 BiLSTM guesses Criminal Code Art. 579 (wrong statute!). A3 gets Art. 4 at Rank 1.

4. Exact Article Term Overlap (BM25 Win)

«តើអ្វីជាការការពារស្របច្បាប់?»

Legitimate self-defense. Expected: Criminal Code Art. 33.

Both BM25 and A3 achieve Rank 1 (100% Top-1) due to clean exact phrase match.